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Beendet am	Montag, 25. September 2023, 18:49
Verbrauchte Zeit	1 Stunde 37 Minuten
Bewertung	48,69 von 100,00

Frage 1

Teilweise richtig

Erreichte Punkte 8,25 von 11,00

Tick the correct statements about the abstract EM-algorithm and the Maximum entropy method.

As in the lecture we use the notation $p(\mathbf{X}, \mathbf{U}, \mathbf{w})$ for a probabilistic model with data distribution $p(\mathbf{X}, \mathbf{w})$ parametrized by $\mathbf{w}$, and latent variables $\mathbf{U}$ distributed according to q . We also use the following decomposition from the lecture: $\ln p(\mathbf{X}, \mathbf{w}) = \mathcal{L}(q, \mathbf{w}) + D_{\text{KL}}(q||p)$. The goal is to maximize this quantity.

Wählen Sie eine oder mehrere Antworten:

- ☒ a. $D_{\text{KL}}(q||p)$ is always greater or equal than 0, and therefore $\mathcal{L}(q, \mathbf{w})$ is a lower bound on the log-likelihood. ✓
- ☒ b. In the E-step, the old parameters $\mathbf{w}_{\text{old}}$ are kept fixed and $\mathcal{L}(q, \mathbf{w}_{\text{old}})$ is minimized with respect to q . ✗
- ☐ c. In the M-step, $\mathcal{L}(q, \mathbf{w})$ is maximized with respect to both q and $\mathbf{w}$.
- ☒ d. After the M-step, $D_{\text{KL}}(q||p)$ becomes strictly positive and $\ln p(\mathbf{X}, \mathbf{w})$ increases. ✓
- ☐ e. The maximum entropy method always assumes a parametrized distribution.
- ☒ f. The maximum entropy method tries to find a suitable distribution based on measurements from data (e.g. mean, standard deviation). ✓
- ☒ g. Entropy is a concept to describe the amount of uncertainty of a distribution. ✓
- ☐ h. The EM-algorithm always converges to the Cramer-Rao bound of $p(\mathbf{X}, \mathbf{w})$.

Frage 2

Falsch

Erreichte Punkte 0,00 von 11,00

Let $w, x \in \mathbb{R}$. Let X be a random variable distributed according to a "supernice" parametrized distribution $p(x, w)$, so that p satisfies all the regularity conditions for any theorem we discussed in the lecture. X shall satisfy $\text{Var}(X) = \frac{1}{w+1}$ and by $I_p(w)$ we denote the associated Fisher information. Moreover, let the following estimators be given:

- An estimator $\hat{w}$ of w so that $E(\hat{w}) = w$.
- For an iid. sample of size l the associated Maximum Likelihood estimator (MLE) $\hat{w}_{ML}(l)$.
- The estimator $\hat{w}_{ML}(\infty) = \lim_{l \rightarrow \infty} \hat{w}_{ML}(l)$

By $\mathcal{N}(\mu, \sigma^2)$ we, as usual, denote the Normal distribution with mean μ and variance σ^2 .

Tick the statements that are true for **any** choice of the terms defined above. In case there are statements about limits, the precise notion of convergence does not matter.

Wählen Sie eine oder mehrere Antworten:

- ☒ a. $\text{Var}(\hat{w}_1) \geq \frac{1}{I_p(w)}$. ✓
- ☒ b. $\text{Var}(X) \geq I_p(w)$. ✗
- ☐ c. $\hat{w}_{ML}(l)$ is unbiased for any l .
- ☒ d. $\hat{w}_{ML}(\infty)$ is distributed according to $\mathcal{N}(I_p(w), \text{Var}(X))$. ✗
- ☐ e. To compute $I_p(w)$, one only requires knowledge of the first derivatives $\frac{\partial p(\cdot, w)}{\partial w}$ of $p(\cdot, w)$.
- ☐ f. The MLE for $E(X)$ for a sample of size l can be computed as $\frac{1}{\hat{w}_{ML}(l)+1}$.
- ☐ g. The variances of $\hat{w}$ and $\hat{w}_{ML}(\infty)$ can be bounded from below simultaneously by the Cramer Rao lower bound.
- ☒ h. $\hat{w}_{ML}(\infty)$ is efficient and unbiased. ✓

Frage 3

Teilweise richtig

Erreichte Punkte 7,56 von 11,00

A random sample X_1, X_2, X_3 of size 3 is drawn iid. from a population with mean $w = 1$ and variance $\sigma^2 = 4$.

Consider the family of estimators $\hat{w}_a = \frac{1}{8}(X_1 + 2aX_2 + X_3)$ for $a \in \mathbb{R}$ of the mean w .

a) Which a makes $\hat{w}_a$ an unbiased estimator?

$a =$ ✗

b) For $a = 1$ compute the bias $b(\hat{w}_1) =$ ✓

c) For $a = 1$ compute the variance $\text{Var}(\hat{w}_1) =$ ✓

d) For $a = 1$ compute the associated MSE :

$\text{MSE}(\hat{w}_1) =$ ✓

IMPORTANT: Only choose an answer if you are sure that it is correct. If you choose no answer, no points will be deducted. If you choose a wrong answer, it will cost you as many points as you would have gotten with the right answer.

Frage 4

Teilweise richtig

Erreichte Punkte 9,78 von 11,00

Imagine you work for the Austrian government and work on a project to gain insights about different agricultural data.

In particular you collected data about the average rainfall, soil quality, bees spread and crop yield of different farmers.

- The farmers' average rainfall is R liters per day.
- The farmers' soil quality is S (a score from 1 to 100).
- The average number of bees humming on the fields is B between 10AM and 4PM (a non-negative real number).
- The farmers' crop yield is C (a score from 1 to 100).

Here are the Structural Equations with noise variables U_R, U_S, U_B, U_C that have been identified by fitting a linear model to the data.

$$R = U_R$$

$$S = 30 + 5R + U_S$$

$$B = 20 + 4R + U_B$$

$$C = 5 + 0.4B + 0.6S + U_C$$

We are also given a specific farm's noise variables:

$$U_R = 9.7$$

$$U_S = 0.9$$

$$U_B = 9.5$$

$$U_C = 4.3$$

a) The experiment's distribution as presented in the SCM is of the following type: ✓

b) Intervening on S and B would make R and C statistically independent: ✗

c) Based on the given noise variables for a specific farm, calculate R, S, B and C.

R: ✓

S: ✓

B: ✓

C: ✓

d) What would be the outcomes R', S', B' and C' if the given farm had made sure that on average 100 bees are humming around on the fields?

R': ✓

S': ✓

B': ✓

C': ✓

Frage 5

Teilweise richtig

Erreichte Punkte 8,25 von 11,00

Let us consider the standard statistical learning theory (SLT) setting from the lecture, i.e.

- Data points $\mathbf{Z} = (\mathbf{x}_i, y_i)_{i=1}^l$ sampled iid. from $p(\mathbf{x}, y)$ supported in $X \times \{-1, 1\}$
- A model class $\mathcal{G}$ of functions $g : X \rightarrow \{-1, 1\}$.
- The zero one loss function L_{z_0} .
- For $g \in \mathcal{G}$ and L_{z_0} the associated generalization error (risk) $R(g)$ and the associated empirical error $R_{\text{emp}}(g, l)$.
- $\mathcal{N}(\mathcal{G}, l)$: the shattering coefficient of $\mathcal{G}$ for a sample of size l .

Tick the correct statements about generalization bounds in this setting. In case there are statements about limits, you do not need to care about the precise notion of convergence.

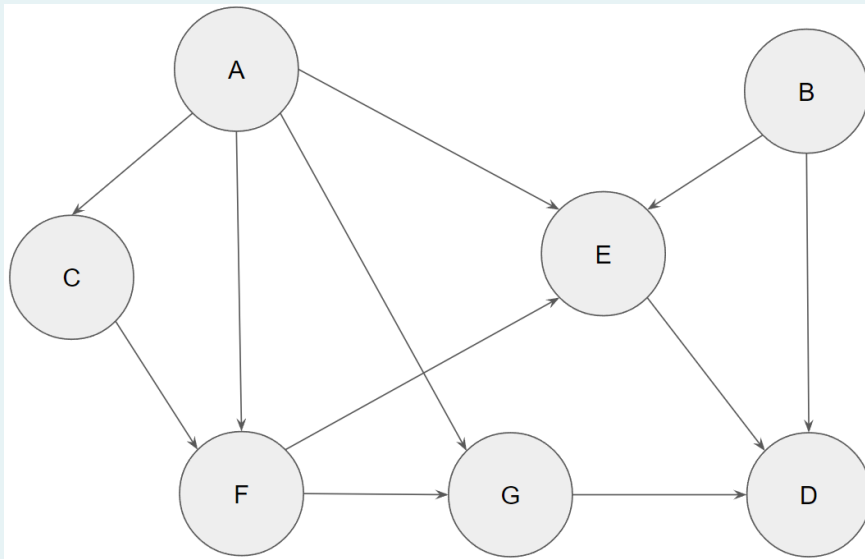
Wählen Sie eine oder mehrere Antworten:

- ☒ a. Let $g \in \mathcal{G}$. The generalization bounds related to the setting described above are of the form $R(g) \leq R_{\text{emp}}(g, l) + \text{capacity term} + \text{confidence term}$. The capacity term measures the size of $\mathcal{G}$, whereas the confidence term accounts for uncertainty. ✓
- ☒ b. Consider the case of a finite model class $\mathcal{G} = \{g_1, \dots, g_m\}$ and fix $\varepsilon > 0$. The union bound allows for lower bounding $p(\sup_{i=1, \dots, m} |R(g_i) - R_{\text{emp}}(g_i, l)| \geq \varepsilon)$ by $\sum_{i=1}^m p(|R(g_i) - R_{\text{emp}}(g_i, l)| \geq \varepsilon)$. ✗
- ☐ c. Fix an error probability $0 < \delta < 1$. The confidence term in the generalization bounds related to the above setting grows with increasing δ and both the capacity and confidence term grow with increasing l .
- ☐ d. Any sample of size l has more than $\mathcal{N}(\mathcal{G}, l)$ realizations/functions in $\mathcal{G}$.
- ☒ e. Let $g \in \mathcal{G}$. The symmetrization lemma allows to upper bound $|R(g) - R_{\text{emp}}(g, l)|$ by a difference $|R_{\text{emp}}(g, l) - R'_{\text{emp}}(g, l)|$ of empirical risks for two iid. samples of sizes l . ✓
- ☒ f. The principle of symmetrization allows to incorporate the shattering coefficient in the capacity term of the generalization bounds. ✓
- ☒ g. For finite VC-dimension $d \leq l$, $\mathcal{N}(\mathcal{G}, l)$ can be upper bounded by a term that is polynomial in l . ✓
- ☐ h. If the VC-dimension of $\mathcal{G}$ is infinite, Sauer's lemma allows us to still get meaningful generalization bounds so that the capacity term decays with increasing l .

Frage 6

Falsch

Erreichte Punkte 0,00 von 12,00

Given the following **Causal Graphical Model**:Tick the boxes if the **corresponding statement is TRUE**.For the **Backdoor-criterion** statements assume the following setup:

- The backdoor-criterion implies that for variables X, Y if we control for the right set of variables Z , the following holds:

$$P(Y|do(X)) = \sum_Z P(Y|X, Z)P(Z)$$
- Tick the "backdoor-criterion" boxes if with the provided sets X, Y, Z fulfill the backdoor criterion according to the formula above.

- ☐ a. The path $A - E - B - D$ is blocked
- ☐ b. The path $A - F - E - B$ is blocked
- ☒ c. The path $F - C - A - G$ is blocked ✖
- ☐ d. Controlling for only G would make F and D independent
- ☐ e. The backdoor-criterion is fulfilled for $X = A, Y = B, Z = \{\}$
- ☒ f. The backdoor-criterion is fulfilled for $X = C, Y = D, Z = \{F\}$ ✖
- ☐ g. The backdoor-criterion is fulfilled for $X = G, Y = C, Z = \{A\}$
- ☐ h. The backdoor-criterion is fulfilled for $X = A, Y = D, Z = \{G, E\}$

Frage 7

Teilweise richtig

Erreichte Punkte 8,25 von 11,00

Let us consider the standard statistical learning theory (SLT) setting from the lecture, i.e.

- Data points $\mathbf{Z} = (\mathbf{x}_i, y_i)_{i=1}^l$ sampled iid. from $p(\mathbf{x}, y)$ supported in $X \times \{-1, 1\}$
- A model class $\mathcal{G}$ of functions $g : X \rightarrow \{-1, 1\}$.
- The zero-one loss function L_{zo} .
- For $g \in \mathcal{G}$ and L_{zo} the associated generalization error (risk) $R(g)$ and the associated empirical error $R_{\text{emp}}(g, l)$.

Tick the correct statements related to this setting.

Wählen Sie eine oder mehrere Antworten:

- ☒ a. Rademacher complexity and VC-dimension are both concepts to describe the size and power of $\mathcal{G}$. ✓
- ☐ b. The computation of the VC-dimension of $\mathcal{G}$ requires to compute an expectation with respect to p .
- ☒ c. The Rademacher complexity intuitively describes how well $\mathcal{G}$ can align with random labellings. These random labellings are modelled by fair coin tosses taking values in $\{-1, 1\}$. ✓
- ☐ d. Rademacher complexity and VC dimension are both concepts to measure the uncertainty in predictions that use models from $\mathcal{G}$.
- ☐ e. It is a general trend that $R_{\text{emp}}(g, l)$ increases, if the capacity term (e.g. VC-dimension or Rademacher complexity) related to $\mathcal{G}$ increases.
- ☒ f. In structural risk minimization, one assumes a nested sequence of function classes $\mathcal{G}_1 \subseteq \mathcal{G}_2 \subseteq \dots$ and selects a good function class and a good function in that class at the same time. ✓
- ☒ g. In regularized risk minimization, instead of only minimizing $R_{\text{emp}}(g, l)$ directly, a term that punishes complex functions is added to the objective. ✓
- ☒ h. Structural risk minimization is a very popular and practical method to control the approximation and estimation error, since it does not need any previous knowledge of the structure of $\mathcal{G}$. ✗

Frage 8

Teilweise richtig

Erreichte Punkte 5,50 von 11,00

Tick the boxes if the corresponding statement is TRUE.

- ☐ a. To control for a set of variables X in Multiple Linear regression one has to add the set of variables X to the set of independent variables. In practice one can simply add all available variables to the set of independent variables.
- ☒ b. To generally solve problems like interventions or counterfactuals, the do-calculus is necessary and sufficient to solve all solvable problems. ✓
- ☐ c. Randomized Control Trials have the advantage of a very simple causal structure. The only problem in practice is, that it's often hard to make a control group.
- ☐ d. Tick true if all of the following 3 statements are correct: (1) Correlation implies Association, (2) Causation does not imply Correlation, (3) Association does not imply Causation
- ☒ e. The identification-estimation procedure describes an abstract algorithm to execute causal inference. It starts with the causal estimand of interest, then uses causal model assumptions and (some instance of) the do-calculus to arrive at a statistical estimand. Given some data, the resulting estimate can be calculated. ✓
- ☐ f. Causality is anchored in physics.
- ☒ g. Sometimes when we model causal relationships, there may be unmeasured causal parents. In general, these are then just omitted from the Causal Graphical Model, since we can't control for them anyways. ✗
- ☒ h. The average causal effect of a treatment T on a drug D is defined as $ACE = \sum_D D \cdot P(D|do(T=1)) - \sum_D D \cdot P(D|do(T=0))$ ✓

Frage 9

Teilweise richtig

Erreichte Punkte 1,10 von 11,00

Consider the data space $X = \mathbb{R}$ and two kernels $k, k' : X \times X \rightarrow \mathbb{R}$ on X . Recall the properties of compositions of kernel functions from the lecture. Recall also the Taylor expansion for the exponential: $\exp(x) = \sum_{i=0}^{\infty} \frac{x^i}{i!}$ valid for any $x \in \mathbb{R}$. Also the following relation may be useful: $\cos(x - y) = \cos(x)\cos(y) + \sin(x)\sin(y)$ for $x, y \in \mathbb{R}$.

Let $x, y \in X$. Tick the functions below, that are again kernels.

Wählen Sie eine oder mehrere Antworten:

- ☒ a. $k_1(x, y) = k(x, y) + 2k'(x, y)$ ✓
- ☒ b. $k_2(x, y) = k(x, y) - 2k'(x, y)$ ✗
- ☐ c. $k_3(x, y) = k(x, y)^2 + 2k'(x, y)$
- ☐ d. $k_4(x, y) = (k(x, y)^2 + 2k'(x, y))^{-2}$
- ☒ e. $k_5(x, y) = \exp(-x)k(x, y)\exp(-y)$ ✓
- ☐ f. $k_6(x, y) = \exp(-x)\exp(k(x, y))\exp(-y)$
- ☒ g. $k_7(x, y) = \cos(x - y)k(x, y)$ ✓